1. $A = \begin{pmatrix} 2 & 2 & 0 & 2 \\ -4 & -4 & -1 & -7 \\ -4 & -4 & -2 & -2 \\ 16 & 16 & 10 & 3 \end{pmatrix}$

$|A| = 2^4 \begin{vmatrix} 1 & 1 & 0 & 1 \\ +4 & +4 & +1 & +7 \\ +4 & +4 & +2 & +2 \\ 16 & 16 & 10 & 3 \end{vmatrix} =$

$= 16(36 - 36 - 336 + 316 + 352 - 352) = 0$

-20 $+20$

2. $\begin{cases} B_1 - B_3 = 0 \\ -2B_1 + B_2 + 3B_3 = -2 \\ -4B_1 + B_2 + 6B_3 = -3 \end{cases}$

$A = \begin{pmatrix} 1 & 0 & -1 \\ -2 & 1 & 3 \\ -4 & 1 & 6 \end{pmatrix}$, $B = \begin{pmatrix} 0 \\ -2 \\ -3 \end{pmatrix}$

$\Delta = 8 - 7 = 1$

$\Delta_1 = \begin{vmatrix} 0 & 0 & -1 \\ -2 & 1 & 3 \\ -3 & 1 & 6 \end{vmatrix} = 2 - 3 = -1$

$\Delta_2 = \begin{vmatrix} 1 & 0 & -1 \\ -2 & -2 & 3 \\ -4 & -3 & 6 \end{vmatrix} = -18 + 17 = -1$

$\Delta_3 = \begin{vmatrix} 1 & 0 & 0 \\ -2 & 1 & -2 \\ -4 & 1 & -3 \end{vmatrix} = -3 + 2 = -1$

$B_1 = \frac{\Delta_1}{\Delta} = -1$; $B_2 = \frac{\Delta_2}{\Delta} = -1$; $B_3 = \frac{\Delta_3}{\Delta} = -1$

3. $-3A + X - 2B = C$

$X = (x_{ij}) = (3a_{ij} + 2b_{ij} + c_{ij})$

$$A = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \\ -1 & 1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 & 3 \\ -2 & 3 & -8 \\ 1 & -1 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 1 & - \\ -1 & 0 & 2 \\ 3 & 1 & -3 \end{pmatrix}$$

$$X = \begin{pmatrix} 6 & -1 & 1 \\ -5 & 9 & -20 \\ 2 & 2 & 3 \end{pmatrix}$$

A B

$$\begin{pmatrix} 2 & 2 & 3 \end{pmatrix}$$

4. $\begin{cases} B_1 - B_2 = -1 \\ B_1 = -1 \\ B_3 = 0 \\ B_2 = 0 \end{cases}$

$$\overbrace{\begin{pmatrix} 1 & -1 & 0 \\ +1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}}^{A} \overbrace{\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}}^{B}$$

$$x = A^{-1} \cdot b = \frac{1}{\det A = 1} \begin{pmatrix} 0 & +1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \begin{cases} B_1 = -1 \\ B_2 = 0 \\ B_3 = 0 \end{cases}$$

5. $\begin{cases} B_1 - B_2 = -1 \\ -2B_1 + 3B_2 - B_3 = 3 \\ -B_1 + B_2 + B_3 = 0 \end{cases}$ $A = \begin{pmatrix} 1 & -1 & 0 \\ -2 & 3 & -1 \\ -1 & 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix}$

$A^{-1} = \frac{1}{\det A} \begin{pmatrix} 4 & 3 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}^T \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix} \Rightarrow \begin{cases} B_1 = -1 \\ B_2 = 0 \\ B_3 = -1 \end{cases}$

$\det A = 2 - 1 = 1$

$\begin{pmatrix} 4 & 1 & 1 \\ 3 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$

6. $\begin{vmatrix} 3 & x & -3 \\ -5 & 3 & x \\ -2 & 5 & -1 \end{vmatrix} = 0$

$-9 - 2x^2 + 75 - 18 - 15x - 5x = 0$

$2x^2 + 20x - 48 = 0$

$x^2 + 10x - 24 = 0$

$x = -12; 2$

7. $\begin{vmatrix} -3 & -3 & 1 & 1 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & -3 \end{vmatrix} = -3 \cdot \begin{vmatrix} -2 & 2 & -2 \\ 0 & 2 & 2 \\ 0 & 0 & -3 \end{vmatrix} = -3 \cdot (-2) \cdot 2 \cdot (-3) = -36$

8. $A = \begin{pmatrix} 7 & -16 & 24 \\ -6 & 17 & -24 \\ -6 & 16 & -23 \end{pmatrix} = (+3)\cdot(-2)\cdot\begin{vmatrix} 2 & +2 \\ 0 & -3 \end{vmatrix} = (-3)\cdot(-2)\cdot 2\cdot(-3) = -36$

$\Rightarrow$

$\Rightarrow A^{-1} = \frac{1}{\det A}\begin{pmatrix} -17\cdot(+23)+16\cdot 24 & A_{21} & A_{31} \\ A_{12} & A_{22} & A_{23} \\ A_{13} & A_{23} & A_{33} \end{pmatrix}$

$= 7\cdot 17\cdot(-23) - 6\cdot 16\cdot 24 - 6\cdot 16\cdot 24 + 6\cdot 17\cdot 24 + 6\cdot 16\cdot 23 + 16\cdot 24 \cdot 7 =$

$= 6\cdot 8 + 7(16\cdot 24 - 17\cdot 23) = 48 - 7(16\cdot 24 - 16\cdot 24 + 17\cdot 24) =$

$= 48 - 49 = -1$

$A^{-1} = -1\cdot\begin{pmatrix} -7 & 6 & 6 \\ -16 & 17 & 16 \\ 24 & -24 & -23 \end{pmatrix}^T = \begin{pmatrix} 7 & -16 & 24 \\ -6 & 17 & -24 \\ -6 & 16 & -23 \end{pmatrix}$

9. $\begin{cases} b_1 - b_2 - b_3 = -1 \\ -b_1 - b_3 = 2 \\ 3b_1 + b_2 + 2b_3 = -6 \end{cases}$

$\left(\begin{array}{ccc|c} 1 & 1 & -1 & -1 \\ -1 & 0 & -1 & 2 \\ 3 & 1 & 2 & -6 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & -1 \\ 3 & 1 & 2 & -6 \\ -1 & 0 & -1 & 2 \end{array}\right)$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & -1 \\ 0 & -2 & 5 & -3 \\ -1 & 0 & -1 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & -1 \\ 0 & -2 & 5 & -3 \\ 0 & 1 & -2 & 1 \end{array}\right) (+2) \sim$$

$$\sim \left(\begin{array}{ccc|c} 1 & 1 & -1 & -1 \\ 0 & 0 & 1 & -1 \\ 0 & 1 & -2 & 1 \end{array}\right) \Rightarrow \begin{cases} b_3 = -1 \\ b_2 = -1 \\ b_1 = -1 \end{cases}$$

10. $A \cdot X \cdot B = C$

$$A = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 \\ -2 & -1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$X = \begin{pmatrix} x_1 & x_3 \\ x_2 & x_4 \end{pmatrix}$$

$$AX = \begin{pmatrix} x_1 - 2x_2 & x_3 - 2x_4 \\ x_1 - x_2 & x_3 - x_4 \end{pmatrix}$$

$$AXB = \begin{pmatrix} x_1 - 2x_2 - 2x_3 + 4x_4 & x_1 - 2x_2 - x_3 + 2x_4 \\ x_1 - x_2 - 2x_3 + 2x_4 & x_1 - x_2 - x_3 + x_4 \end{pmatrix}$$

$$\begin{cases} x_1 - 2x_2 - 2x_3 + 4x_4 = 1 \\ x_1 - 2x_2 - x_3 + 2x_4 = 0 \end{cases} \Rightarrow -x_3 + 2x_4 = 1$$

$$\begin{cases} x_1 - x_2 - 2x_3 + 2x_4 = 1 \\ x_1 - x_2 - x_3 + x_4 = 1 \end{cases} \Rightarrow -x_3 + x_4 = 0$$

$$\Rightarrow x_4 = 1, \; x_3 = 1$$

$$\begin{cases} x_1 - x_2 = 1 \\ x_1 - 2x_2 = -1 \end{cases} \Rightarrow x_2 = +2; \; x_1 = 3$$

$$X = \begin{pmatrix} +3 & 1 \\ +2 & 1 \end{pmatrix}$$

11. $\begin{pmatrix} -2 & 1 & -3 \\ -2 & -4 & -1 \\ -5 & 4 & 3 \end{pmatrix} \cdot \begin{pmatrix} -2 & 2 & 1 \\ 0 & -2 & 2 \\ 2 & 2 & 2 \end{pmatrix}^T - 1 \cdot \begin{pmatrix} 1 & 3 & +2 \\ -3 & -1 & 3 \\ -5 & -2 & -2 \end{pmatrix} =$

$= \begin{pmatrix} -2 & 1 & -3 \\ -2 & -4 & -1 \\ -5 & 4 & 3 \end{pmatrix} \begin{pmatrix} -2 & 0 & 2 \\ 2 & -2 & 2 \\ 1 & 2 & 2 \end{pmatrix} + \begin{pmatrix} -1 & -3 & -2 \\ 3 & 1 & -3 \\ 5 & 2 & 2 \end{pmatrix} =$

$$= \begin{pmatrix} 3 & -8 & -8 \\ -5 & 6 & -14 \\ 21 & -2 & 4 \end{pmatrix} + \begin{pmatrix} -1 & -3 & -2 \\ 3 & 1 & -3 \\ 5 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -11 & -10 \\ -2 & 7 & -17 \\ 26 & 0 & 6 \end{pmatrix}$$

12. $-5A - 5X + 20B = C$

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & -1 & 1 \\ -1 & 2 & -3 \\ 0 & 0 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} -5 & 0 & 10 \\ 10 & -5 & -25 \\ -5 & 0 & 5 \end{pmatrix}$$

$$X = (x_{ij}) = \left(\frac{-5a_{ij} + 20b_{ij} - c_{ij}}{5}\right) =$$

$$= (-a_{ij} + 4b_{ij} - d_{ij}) = \begin{pmatrix} 4 & -2 & -1 \\ -6 & 8 & -5 \\ 1 & 0 & 2 \end{pmatrix}$$

$$D = d_{ij} = \left(\frac{c_{ij}}{5}\right) = \begin{pmatrix} -1 & 0 & 2 \\ 2 & -1 & -5 \\ -1 & 0 & 1 \end{pmatrix}$$

$$\begin{vmatrix} 0 & -2 & -5 & -1 \\ 0 & 0 & 4 & -3 \\ 0 & 3 & 0 & -3 \\ 4 & -5 & 2 & 0 \end{vmatrix} = -4 \cdot \begin{vmatrix} -2 & -5 & -1 \\ 0 & 4 & -3 \\ 3 & 0 & -3 \end{vmatrix} = -324$$

14. $A = \begin{pmatrix} 1 & -1 & 1 & 2 \\ -1 & 2 & -1 & -3 \\ 2 & -4 & 3 & 6 \\ 3 & -6 & 4 & 10 \end{pmatrix}$

$\alpha_{23} = (-1)^5 M_{23} = -M_{23}$

$M_{23} = \begin{vmatrix} 1 & -1 & 2 \\ 2 & -4 & 6 \\ 3 & -6 & 10 \end{vmatrix} = -82 + 24 + 20 + 36 = -2$

$\alpha_{23} = 2$